Hours worked and earnings inequality over time

2025 SEA Meeting

Tyler Schumacher

November 21, 2025

Outline

Definitions: Hours, Wages, and Earnings

- I focus on the relationship of *weekly* hours to *weekly* earnings
  - *Weekly Earnings* $=$ *Weekly Hours* $\times$ *Hourly wage*

Definitions: Hours, Wages, and Earnings

- I focus on the relationship of *weekly* hours to *weekly* earnings
  - $\textit{Weekly Earnings} = \textit{Weekly Hours} \times \textit{Hourly wage}$
- This framing is used for both hourly and salaried workers
  - For salaried workers, "*Wage*" refers to $\frac{\textit{Earnings}}{\textit{Hours}}$

Motivation: Rising earnings inequality

- Earnings inequality has increased over last half century (see, e.g., Heathcote et al, 2023)
  - Variance of log earnings, 90/10 percentile ratio, 75/25 percentile ratio have all increased

Motivation: Rising earnings inequality

- Earnings inequality has increased over last half century (see, e.g., Heathcote et al, 2023)
  - Variance of log earnings, 90/10 percentile ratio, 75/25 percentile ratio have all increased

- One explanation: "skill biased technological change" (*e.g.*, Autor et al, 2003; Goos et al, 2014; Acemoglu & Autor, 2011)
  - Computers substitute for "routine" work, ↑ skill premium

Motivation: Rising earnings inequality

- Earnings inequality has increased over last half century (see, e.g., Heathcote et al, 2023)
  - Variance of log earnings, 90/10 percentile ratio, 75/25 percentile ratio have all increased
- One explanation: "skill biased technological change" (*e.g.*, Autor et al, 2003; Goos et al, 2014; Acemoglu & Autor, 2011)
  - Computers substitute for "routine" work, ↑ skill premium
- *Vast* majority of literature focuses on **wage** inequality as driver
  - Remember that $\textit{Earnings} = \textit{Hours} \times \textit{Wage}$

Motivation: Rising earnings inequality

- Earnings inequality has increased over last half century (see, e.g., Heathcote et al, 2023)
    - Variance of log earnings, 90/10 percentile ratio, 75/25 percentile ratio have all increased
- One explanation: "skill biased technological change" (*e.g.*, Autor et al, 2003; Goos et al, 2014; Acemoglu & Autor, 2011)
    - Computers substitute for "routine" work, ↑ skill premium
- *Vast* majority of literature focuses on **wage** inequality as driver
    - Remember that $Earnings = Hours \times Wage$
    - There is little change in **hours** inequality as measured by, *e.g.*, variance of log hours

Motivation: Rising earnings inequality

- Earnings inequality has increased over last half century (see, e.g., Heathcote et al, 2023)
  - Variance of log earnings, 90/10 percentile ratio, 75/25 percentile ratio have all increased
- One explanation: "skill biased technological change" (*e.g.*, Autor et al, 2003; Goos et al, 2014; Acemoglu & Autor, 2011)
  - Computers substitute for "routine" work, $\uparrow$ skill premium
- *Vast* majority of literature focuses on **wage** inequality as driver
  - Remember that $Earnings = Hours \times Wage$
  - There is little change in **hours** inequality as measured by, *e.g.*, variance of log hours
- However, it also matters **who** is working the longer hours
  - Precisely, $cov(wages, hours)$ contributes to earnings inequality

This project: Empirical contribution

- Roughly 20% of rise in earnings inequality since 1975 among working men is accounted for by increase in $cov(wages, hours)$

This project: Empirical contribution

- Roughly 20% of rise in earnings inequality since 1975 among working men is accounted for by increase in $cov(wages, hours)$
  - $cov(wages, hours) \uparrow$ from slightly negative to slightly positive

This project: Empirical contribution

- Roughly 20% of rise in earnings inequality since 1975 among working men is accounted for by increase in $cov(wages, hours)$
  - $cov(wages, hours) \uparrow$ from slightly negative to slightly positive
- Substantial portion of change is between-occupation

This project: Empirical contribution

- Roughly 20% of rise in earnings inequality since 1975 among working men is accounted for by increase in $cov(wages, hours)$
  - $cov(wages, hours) \uparrow$ from slightly negative to slightly positive
- Substantial portion of change is between-occupation
- Hours by occupation stable over time, high-hour occupations had higher wage growth

This project: Empirical contribution

- Roughly 20% of rise in earnings inequality since 1975 among working men is accounted for by increase in $cov(wages, hours)$
  - $cov(wages, hours) \uparrow$ from slightly negative to slightly positive
- Substantial portion of change is between-occupation
- Hours by occupation stable over time, high-hour occupations had higher wage growth
- High occupational hours related to task content of occupations

This project: Theoretical contribution

- Question: Why has the relationship between hours and wages changed over time?

This project: Theoretical contribution

- Question: Why has the relationship between hours and wages changed over time?
- Set forth model rationalizing hours differences by occupation
  - Workers work long hours if it is difficult to pass off task-specific knowledge to other employee

This project: Theoretical contribution

- Question: Why has the relationship between hours and wages changed over time?
- Set forth model rationalizing hours differences by occupation
  - Workers work long hours if it is difficult to pass off task-specific knowledge to other employee
- Adds hours to skill-biased technological change literature
  - "Hours-biased technological change" *complements*, does *not* contradict, SBTC

This project: Theoretical contribution

- Question: Why has the relationship between hours and wages changed over time?
- Set forth model rationalizing hours differences by occupation
  - Workers work long hours if it is difficult to pass off task-specific knowledge to other employee
- Adds hours to skill-biased technological change literature
  - "Hours-biased technological change" *complements*, does *not* contradict, SBTC
- Short-hour occupations more "routine", $\therefore$ relative wage $\downarrow$ from computers

Outline

Data

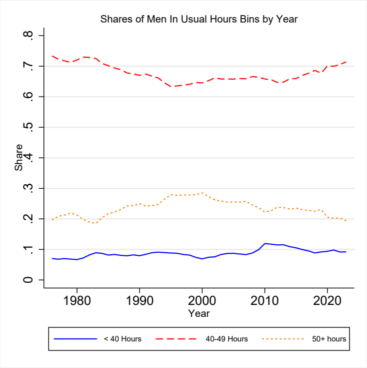
- CPS ASEC data from 1975 - 2023
- Main variables:
  - Usual hours worked per week last year
  - Weekly earnings ($\frac{\text{earnings}}{\text{weeks worked last year}}$)
  - Education level
  - Age

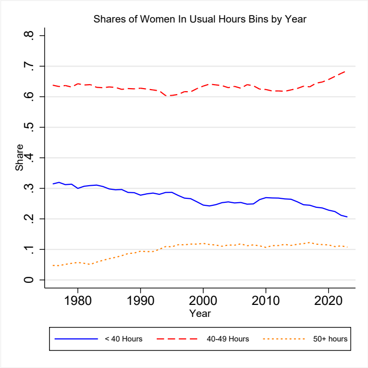
Data

- CPS ASEC data from 1975 - 2023
- Main variables:
  - Usual hours worked per week last year
  - Weekly earnings ($\frac{\text{earnings}}{\text{weeks worked last year}}$)
  - Education level
  - Age
- Main sample criteria:
  - worked at least 20 usual weekly hours
  - implied wage of at least half federal minimum wage
  - Ages 25-64

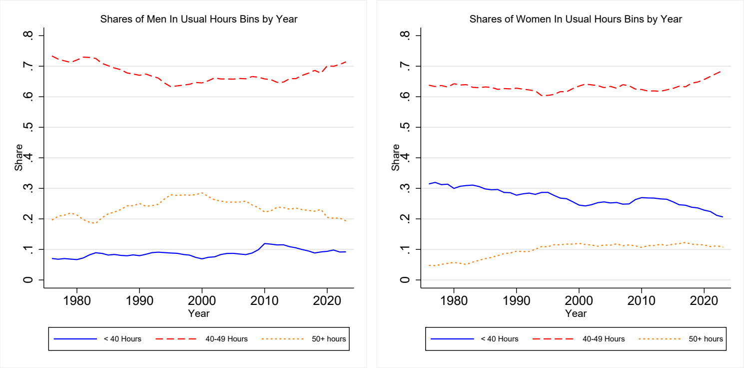
Topcoding

Shares in hours bins (short, medium, long) by year





Shares in hours bins (short, medium, long) by year



- There IS variation in weekly hours worked!

Example Histograms

Variance Decomposition of Log Earnings: Approach

- Let $\tilde{e} = \ln(\textit{weekly earnings})$, $\tilde{w} = \ln(\textit{wage})$, $\tilde{n} = \ln(\textit{weekly hours})$

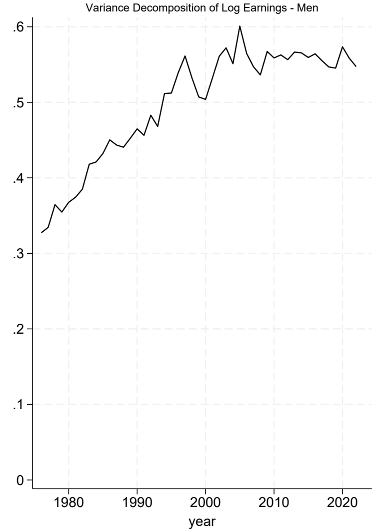
Variance Decomposition of Log Earnings: Approach

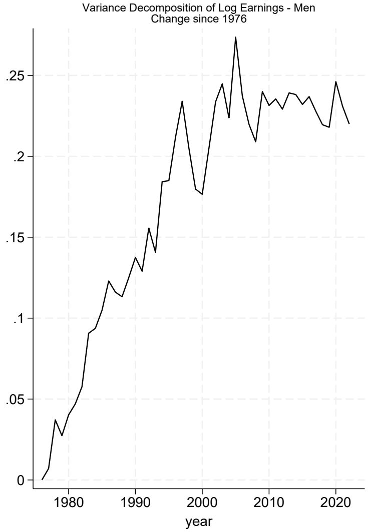
- Let $\tilde{e} = \ln(\textit{weekly earnings})$, $\tilde{w} = \ln(\textit{wage})$, $\tilde{n} = \ln(\textit{weekly hours})$
- $\textit{Earnings} = \textit{Hours} \times \textit{Wages}$, so $\tilde{e} = \tilde{w} + \tilde{n}$

Variance Decomposition of Log Earnings: Approach

- Let $\tilde{e} = \ln(\textit{weekly earnings})$, $\tilde{w} = \ln(\textit{wage})$, $\tilde{n} = \ln(\textit{weekly hours})$
- $\textit{Earnings} = \textit{Hours} \times \textit{Wages}$, so $\tilde{e} = \tilde{w} + \tilde{n}$
- So, $var(\tilde{e}) = \mathrm{var}(\tilde{w}) + \mathrm{var}(\tilde{n}) + 2\,\mathrm{cov}(\tilde{w}, \tilde{n})$

Variance Decomposition of Log Earnings



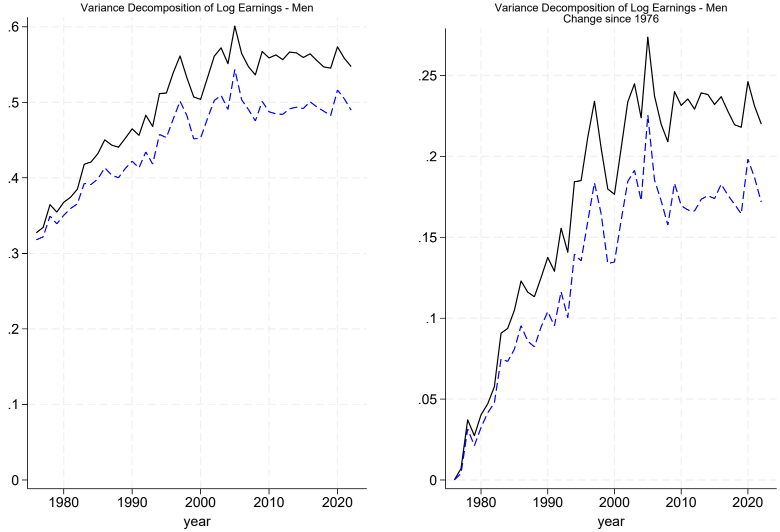


$\text{var}(\tilde{e}) =$

Earnings-hours elasticity approach

Measurement error

Variance Decomposition of Log Earnings

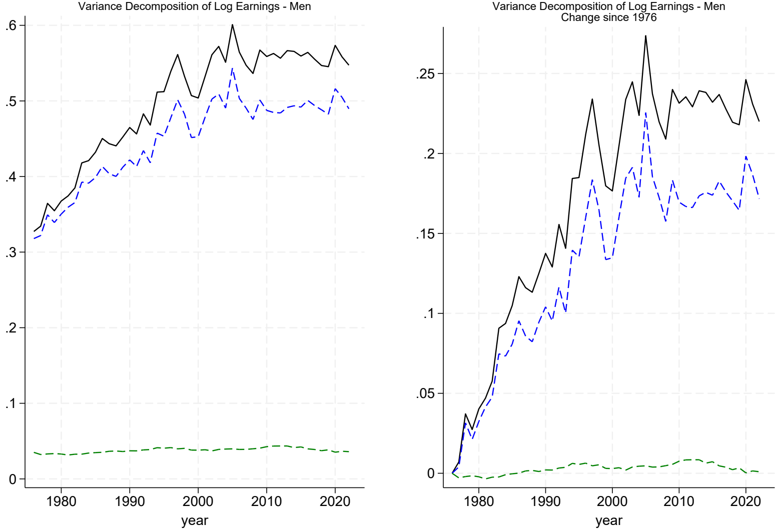


$\text{var}(\tilde{e}) = \text{var}(\tilde{w}) +$

Earnings-hours elasticity approach

Measurement error

Variance Decomposition of Log Earnings

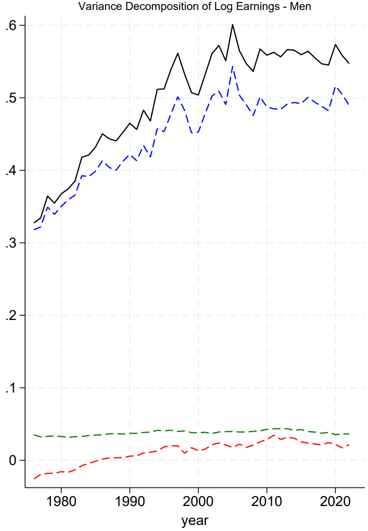


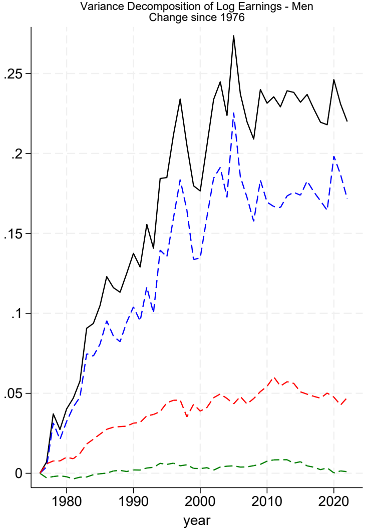
$\mathrm{var}(\tilde{e}) = \mathrm{var}(\tilde{w}) + \mathrm{var}(\tilde{n}) +$

Earnings-hours elasticity approach

Measurement error

Variance Decomposition of Log Earnings





$$\mathrm{var}(\tilde{e}) = \mathrm{var}(\tilde{w}) + \mathrm{var}(\tilde{n}) + 2\,\mathrm{cov}(\tilde{w}, \tilde{n})$$

Earnings-hours elasticity approach

Measurement error

Shares accounted for in variance decomposition

Measure	Change over full period	Share of Change in $\tilde{e}$
$\text{var}(\tilde{e})$	0.2200	100%
$\text{var}(\tilde{w})$	0.1716	78.0%
$\text{var}(\tilde{n})$	0.0009	0.4%
$2\,\text{cov}(\tilde{w}, \tilde{n})$	0.0473	21.5%

Earnings-hours elasticity approach Measurement error

Robustness of covariance result

- "Earnings-hours elasticity" approach Approach
  - Counterfactual exercises using elasticity Counterfactuals
- Robustness of elasticities to basic controls (education, age) Controls
- Different nationally representative datasets Different datasets

The role of occupations

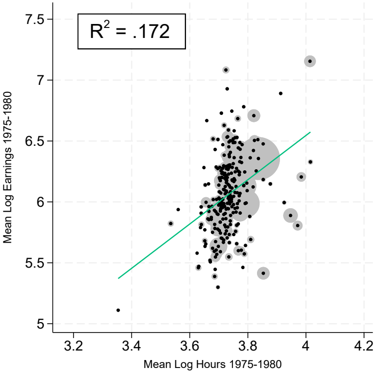
- We can further decompose into between- and within-occupation variance components
- Consistent occupations over time with "occ1990dd" classifications from Autor & Dorn, 2013
- By Law of Total Covariance, $cov(\hat{w}, \hat{n})$ can be decomposed as:

$$\underbrace{\mathbb{E}[cov(\hat{w}, \hat{n}|occ = o)]}_{\text{``within'' component}} + \underbrace{cov(\mathbb{E}[\hat{w}|occ = o], \mathbb{E}[\hat{n}|occ = o])}_{\text{``between'' component}}$$

Variance Decomposition of Log Earnings

- Between-occupation changes account for over half of change by late 2000s

Occupation-level hours-earnings relationship: 1970s and 2000s



- Hours-wage relationship strengthens from 1970s to 2000s

Explanations for between-occupation covariance

- Increasing between-occupation covariance could come from:
  1. High-hour occupations grow in wage
  2. High-wage occupations grow in hours
  3. Hours and wages grow together in some occupations
  4. Movement of workers to high-hour, high-wage occupations

Explanations for between-occupation covariance

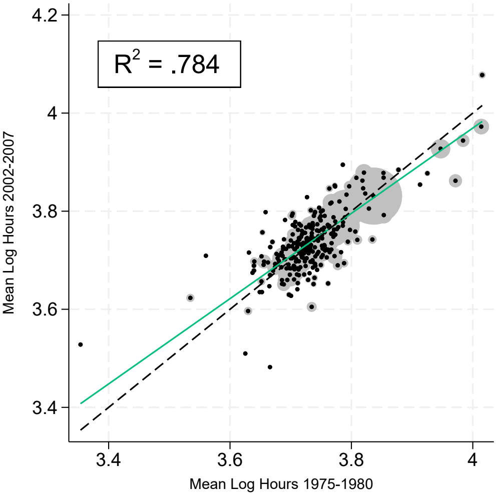
- Increasing between-occupation covariance could come from:
    1. High-hour occupations grow in wage
    2. High-wage occupations grow in hours
    3. Hours and wages grow together in some occupations
    4. Movement of workers to high-hour, high-wage occupations

- Here: demonstrate relevance of #2

Explanations for between-occupation covariance

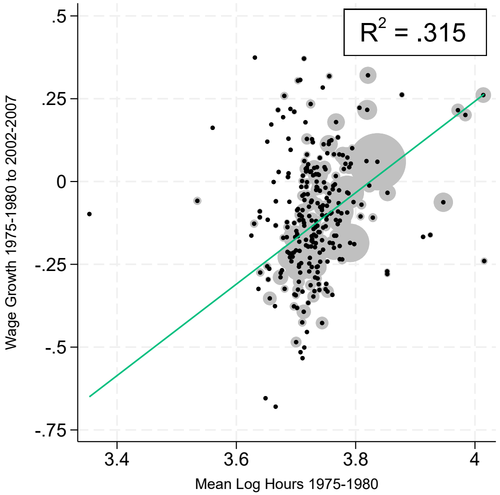
- Increasing between-occupation covariance could come from:
  1. High-hour occupations grow in wage
  2. High-wage occupations grow in hours
  3. Hours and wages grow together in some occupations
  4. Movement of workers to high-hour, high-wage occupations
- Here: demonstrate relevance of #2
- Take 1975-1980 as "early" period, 2002-2007 as "late" period

"Early" hours and "late" hours



- Hours change very little at occupation level

Hours and wage growth



- High-hour occupations had higher wage growth

Hours growth and wage growth | Hours growth and occupation shares

Some key concepts for occupation analysis

- "Handoff-ability": the extent to which one worker can end a work spell and another can take over
  - "Handoff-able" examples: cashiers, janitors, data entry
  - Non "handoff-able" examples: creative directors, senior managers, research scientists

Some key concepts for occupation analysis

- "Handoff-ability": the extent to which one worker can end a work spell and another can take over
  - "Handoff-able" examples: cashiers, janitors, data entry
  - Non "handoff-able" examples: creative directors, senior managers, research scientists
- Important: *conditional on both workers being trained and capable*
  - Not about whether worker can be replaced by a random person, but that a coworker can step in

Some key concepts for occupation analysis

- "Handoff-ability": the extent to which one worker can end a work spell and another can take over
  - "Handoff-able" examples: cashiers, janitors, data entry
  - Non "handoff-able" examples: creative directors, senior managers, research scientists
- Important: *conditional on both workers being trained and capable*
  - Not about whether worker can be replaced by a random person, but that a coworker can step in
- Types of task content following Autor et al., 2003 Autor & Dorn, 2013
  - *Routine* work activities follow well-defined, rules-based procedures and are easily automated
  - *Abstract* work activities require problem-solving, analysis, creativity, and complex communication
  - *Manual* tasks involve bodily effort, manipulation, or movement

Measuring handoff-ability

- Handoff-ability measure not readily available in surveys

Measuring handoff-ability

- Handoff-ability measure not readily available in surveys
- This paper: cross-validate between new measures

Measuring handoff-ability

- Handoff-ability measure not readily available in surveys
- This paper: cross-validate between new measures
- First: generate measures using two LLMs (ChatGPT, Claude) Details
  - Similar in spirit to Eloundou et al, 2024's influential work

Measuring handoff-ability

- Handoff-ability measure not readily available in surveys
- This paper: cross-validate between new measures
- First: generate measures using two LLMs (ChatGPT, Claude) Details
  - Similar in spirit to Eloundou et al, 2024's influential work
- Then, validate against novel items in the Real-Time Population Survey (RPS) (see Bick & Blandin, 2023)

Measuring handoff-ability

- Handoff-ability measure not readily available in surveys
- This paper: cross-validate between new measures
- First: generate measures using two LLMs (ChatGPT, Claude) Details
  - Similar in spirit to Eloundou et al, 2024's influential work
- Then, validate against novel items in the Real-Time Population Survey (RPS) (see Bick & Blandin, 2023)
- I also argue that handoff-ability is highly related to *abstractness*, which Autor & Dorn, 2013 measure

Validation: Real-Time Population Survey (RPS)

- Novel items were included in Aug. 2025 wave of the RPS:

Validation: Real-Time Population Survey (RPS)

- Novel items were included in Aug. 2025 wave of the RPS:
- *How often are tasks at work dependent on you personally?*
  - Never, Less than half the time, About half the time, More than half the time, Always or almost always

Validation: Real-Time Population Survey (RPS)

- Novel items were included in Aug. 2025 wave of the RPS:
- *How often are tasks at work dependent on you personally?*
  - Never, Less than half the time, About half the time, More than half the time, Always or almost always
- *When you stop working for the day what happens?*
  - My work pauses until I return
  - Someone else takes over my position or tasks

Validation: Real-Time Population Survey (RPS)

- Novel items were included in Aug. 2025 wave of the RPS:
- *How often are tasks at work dependent on you personally?*
  - Never, Less than half the time, About half the time, More than half the time, Always or almost always
- *When you stop working for the day what happens?*
  - My work pauses until I return
  - Someone else takes over my position or tasks
- *In your job, how important is knowledge that is difficult to write down or explain – such as personal judgment or intuition – for doing your tasks effectively?*
  - Very important, moderately important, somewhat important, not at all important

RPS Cross-Validation

- People in more LLM-measured “handoff-able” jobs report that...
  - tasks are more dependent on them personally
  - their work is more likely to pause until they return when they stop working
  - they rely more on personal judgment or intuition

Outline

Overview

- Some jobs short-hour and others long-hour, stable over time
- Higher relative wage growth in long-hour jobs over time
- Goals of model:
  - Why are some jobs long-hour at all times?

Overview

- Some jobs short-hour and others long-hour, stable over time
- Higher relative wage growth in long-hour jobs over time
- Goals of model:
  - Why are some jobs long-hour at all times?
  - What makes those jobs grow more in wage?

Overview

- Some jobs short-hour and others long-hour, stable over time
- Higher relative wage growth in long-hour jobs over time
- Goals of model:
  - Why are some jobs long-hour at all times?
  - What makes those jobs grow more in wage?
- **Today:** *highly* stylized model P.E. model showing key novel mechanism

Overview

- Some jobs short-hour and others long-hour, stable over time
- Higher relative wage growth in long-hour jobs over time
- Goals of model:
  - Why are some jobs long-hour at all times?
  - What makes those jobs grow more in wage?
- **Today:** *highly* stylized model P.E. model showing key novel mechanism
- **Full paper**: G.E. model with worker heterogeneity and sorting, endogenous computer adoption, labor reallocation

Model intuition

- Imagine an employer wants 80 hours of work done

Model intuition

- Imagine an employer wants 80 hours of work done
- They could hire 2 employees each working 40 hours

Model intuition

- Imagine an employer wants 80 hours of work done
- They could hire 2 employees each working 40 hours
- Could also hire 1 person to work 80 hours

Model intuition

- Imagine an employer wants 80 hours of work done
- They could hire 2 employees each working 40 hours
- Could also hire 1 person to work 80 hours
  - People don't like working 80 hours → requires pay premium

Model intuition

- Imagine an employer wants 80 hours of work done
- They could hire 2 employees each working 40 hours
- Could also hire 1 person to work 80 hours
  - People don't like working 80 hours → requires pay premium
- If two employees, "handoffs" of work are required

Model intuition

- Imagine an employer wants 80 hours of work done
- They could hire 2 employees each working 40 hours
- Could also hire 1 person to work 80 hours
  - People don't like working 80 hours → requires pay premium
- If two employees, "handoffs" of work are required
  - Handoffs are a costly process as workers have to coordinate
  - Handoffs easier in some occupations (*e.g.* cashiers), harder in others (*e.g.* doctors)

Model intuition

- Imagine an employer wants 80 hours of work done
- They could hire 2 employees each working 40 hours
- Could also hire 1 person to work 80 hours
    - People don't like working 80 hours → requires pay premium
- If two employees, "handoffs" of work are required
    - Handoffs are a costly process as workers have to coordinate
    - Handoffs easier in some occupations (*e.g.* cashiers), harder in others (*e.g.* doctors)
- Firm tradeoff: pay for long hours or incur handoff costs

Simplified reduced-form model: Words

- Occupations j differ in *handoff frictions*, measured as hours spent handing off work

Simplified reduced-form model: Words

- Occupations j differ in *handoff frictions*, measured as hours spent handing off work
- Firm faces a convex wage bill in hours (see, *e.g.*, Goldin, 2014)
    - Can be microfounded with convex labor disutility for worker

Simplified reduced-form model: Words

- Occupations j differ in *handoff frictions*, measured as hours spent handing off work
- Firm faces a convex wage bill in hours (see, *e.g.*, Goldin, 2014)
  - Can be microfounded with convex labor disutility for worker
- Firm chooses number of workers m, hours per worker H to produce given quantity L

Simplified reduced-form model: Words

- Occupations j differ in *handoff frictions*, measured as hours spent handing off work
- Firm faces a convex wage bill in hours (see, *e.g.*, Goldin, 2014)
  - Can be microfounded with convex labor disutility for worker
- Firm chooses number of workers m, hours per worker H to produce given quantity L
- Firm pays workers for handoff hours η_j, but no production during them

Simplified reduced-form model: Words

- Occupations j differ in *handoff frictions*, measured as hours spent handing off work
- Firm faces a convex wage bill in hours (see, *e.g.*, Goldin, 2014)
  - Can be microfounded with convex labor disutility for worker
- Firm chooses number of workers m, hours per worker H to produce given quantity L
- Firm pays workers for handoff hours η_j, but no production during them
- Workers are identical, assigned to occupation j

Simplified reduced-form model: Math

- Firm solves a cost minimization problem per amount L produced:

$$\min_{H \geqslant \eta(T), m \geqslant 0} m \cdot \kappa_j H^{1+\alpha}$$

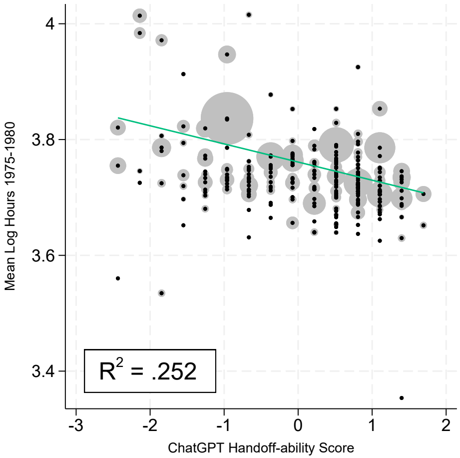
$$\text{s.t. } L = m[H - \eta_j]$$

- Parameters
  - α: degree of contract convexity
  - η_j: handoff frictions
  - κ_j: occupation wage shifter
- Choice variables
  - m: number of workers
  - H: hours per worker

Firm solution

- $H^* = \eta_j \left(1 + \frac{1}{\alpha}\right)$
  - Optimal hours per worker increasing in η_j
  - Interpretation: less handoff-able job $\rightarrow$ longer hours
- $m^* = \frac{L}{\eta_j\left(1+\frac{1}{\alpha}\right)}$
  - To produce more ($L \uparrow$), firms scale *headcount*, not hours

Hours and Handoff-ability



- Hours decreasing in handoff-ability

Claude scores | Binscatter | Large hours values trimmed

When do computers increase hours-wage correlation?

- Computers increase $cov(\hat{H}^*, \hat{w}^*)$ if more handoff-able occupations hurt by computers

When do computers increase hours-wage correlation?

- Computers increase $cov(\hat{H}^*, \hat{w}^*)$ if more handoff-able occupations hurt by computers
  - Autor et al (2003): "hurt by computers" $\leftrightarrow$ "routine"

When do computers increase hours-wage correlation?

- Computers increase $cov(\hat{H}^*, \hat{w}^*)$ if more handoff-able occupations hurt by computers
  - Autor et al (2003): "hurt by computers" $\leftrightarrow$ "routine"
  - **Intuition:** Repetitive, rules-based tasks can be taught to computers *and* handed off between humans

When do computers increase hours-wage correlation?

- Computers increase $cov(\hat{H}^*, \hat{w}^*)$ if more handoff-able occupations hurt by computers
  - Autor et al (2003): "hurt by computers" $\leftrightarrow$ "routine"
  - **Intuition:** Repetitive, rules-based tasks can be taught to computers *and* handed off between humans
- This can be tested with our proxies

When do computers increase hours-wage correlation?

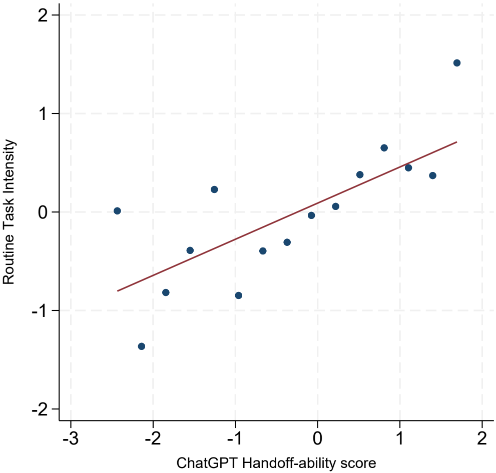
- Computers increase $cov(\hat{H}^*, \hat{w}^*)$ if more handoff-able occupations hurt by computers
  - Autor et al (2003): "hurt by computers" ↔ "routine"
  - **Intuition:** Repetitive, rules-based tasks can be taught to computers *and* handed off between humans

- This can be tested with our proxies

- Define *Routine Task Intensity* (RTI_j) as in Autor & Dorn (2013):

$$RTI_j = \ln(\text{routine input}) - \ln(\text{abstract input}) - \ln(\text{manual input})$$

Routine task intensity and handoff-ability



Coeff = .367 (t = 3.4). R^2 = .128

- Handoff-able jobs are more routine

Outline

Conclusions so far

- For men, the cross-sectional elasticity of earnings with respect to hours has increased since 1976

Conclusions so far

- For men, the cross-sectional elasticity of earnings with respect to hours has increased since 1976
- Relatedly, the observed correlation of log wages with log hours has changed from negative to positive

Conclusions so far

- For men, the cross-sectional elasticity of earnings with respect to hours has increased since 1976
- Relatedly, the observed correlation of log wages with log hours has changed from negative to positive
- A simple occupation model suggests a relationship between ease of "handoffs" and lower hours

Conclusions so far

- For men, the cross-sectional elasticity of earnings with respect to hours has increased since 1976
- Relatedly, the observed correlation of log wages with log hours has changed from negative to positive
- A simple occupation model suggests a relationship between ease of "handoffs" and lower hours
- Computers hurt workers in routine occupations, which are more likely to be handoff-able

Conclusions so far

- For men, the cross-sectional elasticity of earnings with respect to hours has increased since 1976
- Relatedly, the observed correlation of log wages with log hours has changed from negative to positive
- A simple occupation model suggests a relationship between ease of “handoffs” and lower hours
- Computers hurt workers in routine occupations, which are more likely to be handoff-able
- Future: more complete calibration of model

Outline

Introduction and Motivation

Empirical Facts

A simple model of hours

Conclusion and future directions

Appendix

Causal relationship between hours and wage

- Economic literature has long acknowledged the existence of a part-time wage penalty (*e.g* Manning & Petrongolo, 2005)

Causal relationship between hours and wage

- Economic literature has long acknowledged the existence of a part-time wage penalty (*e.g* Manning & Petrongolo, 2005)
- Recent literature (*e.g* Bick, Blandin, & Rogerson, 2022) has also established a "long hours" wage penalty

Causal relationship between hours and wage

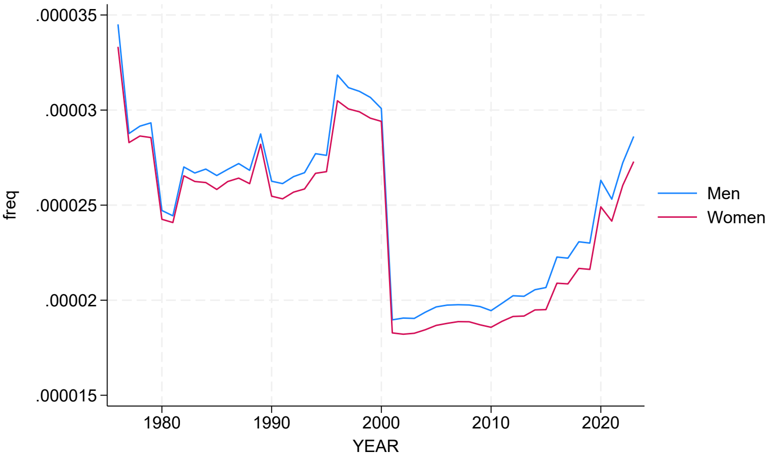
- Economic literature has long acknowledged the existence of a part-time wage penalty (*e.g* Manning & Petrongolo, 2005)
- Recent literature (*e.g* Bick, Blandin, & Rogerson, 2022) has also established a "long hours" wage penalty
- Hourly pay peaks at roughly 45 hours per week, declines for higher and lower hours

Causal relationship between hours and wage

- Economic literature has long acknowledged the existence of a part-time wage penalty (*e.g* Manning & Petrongolo, 2005)
- Recent literature (*e.g* Bick, Blandin, & Rogerson, 2022) has also established a "long hours" wage penalty
- Hourly pay peaks at roughly 45 hours per week, declines for higher and lower hours
- This relationship is widely accepted as causal, where part-time work induces lower hourly pay

Back to intro

Frequencies Topcoded by Year



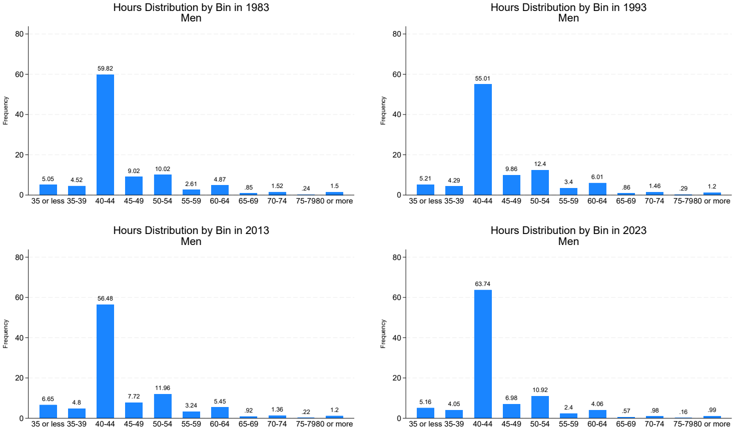
Back to Data

Topcoding Adjustment

- I follow (*e.g* Heathcote et al, 2023)
- Intuition: Assume distribution continues smoothly after topcoding threshold, increase topcoded values to mean of forecasted distribution
- Procedure: Assume underlying income distribution is Pareto
  - Estimate parameters of Pareto distribution from income values near the topcoding threshold
  - Then, forecast mean value of topcoded values using these estimates

Back to Data

Sample distributions of hours worked



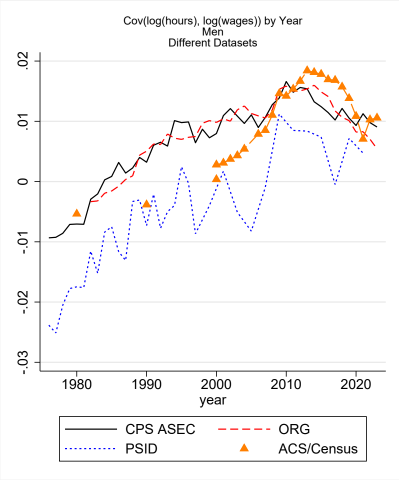
Back to graph

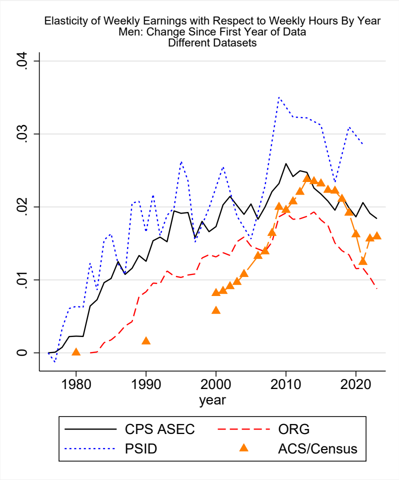
Measurement error: Explanation

- Concern: hours are measured with error
- Classic problem of "division bias" (*e.g* Borjas , 1980)
  - $cov(\ln(hrs), \ln(wage))$ is biased *downward* by division bias
- Key result here: elasticity has *increased over time*
  - Measurement error could explain result if measurement error *decreased over time*
  - Problem if, *e.g.*, CPS-ASEC's hours questions got better over time
- One simple check: do we see similar rise in other datasets?
- Other datasets with sufficient info: ACS/Census, PSID, CPS-ORG

back

Results from other datasets





back

Another method: Earnings-hours elasticities

- This is directly related to the cross-sectional relationship between hours and earnings
- Looking at relationship as an elasticity allows us to:
  - Add basic controls (*e.g.* education, age)
  - Examine long- and short-hour elasticities (see, *e.g.*, Bick et al, 2022)

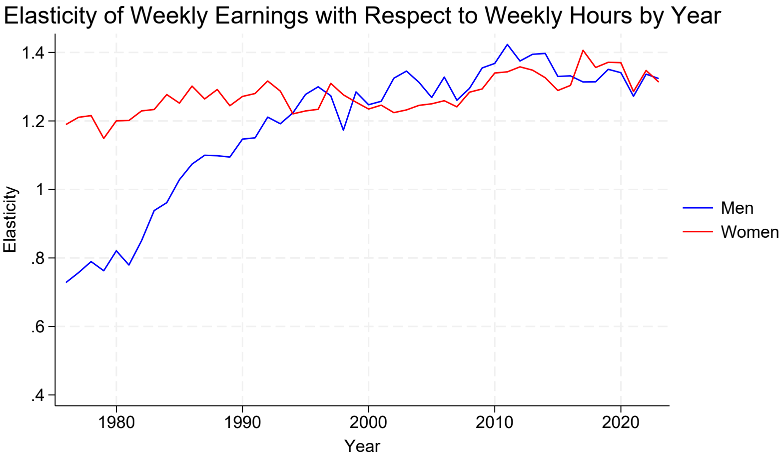
Elasticities of earnings with respect to hours: Approach

- Approach: Find earnings-hours elasticity by year
  - Get separate estimates for men and women
- Estimate the following:
- $\ln(Earnings) = \alpha + \varepsilon \ln(\frac{hours}{40}) + \gamma X$

  where X is control vector where applicable, ε is elasticity coefficient, hours and earnings are at weekly level

back

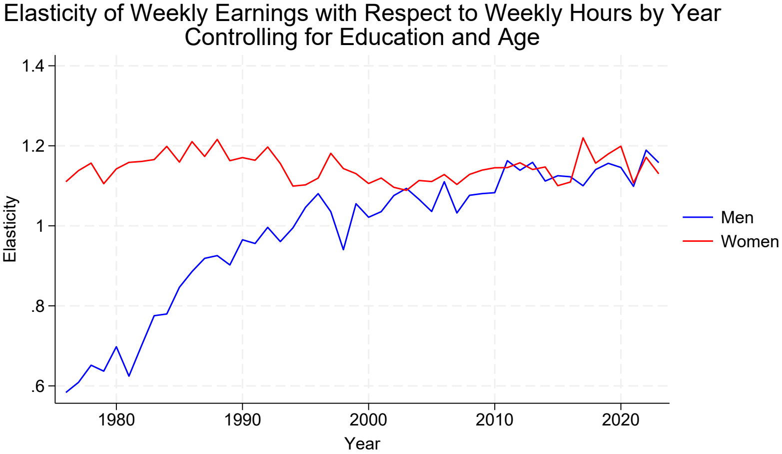
Elasticities of earnings with respect to hours



- Elasticity increased for men from 1975 to 2000s, was relatively constant for women

Separate long- and short-hour elasticities

Elasticities, Controlling for education and age



- Pattern remains when controlling for education and age

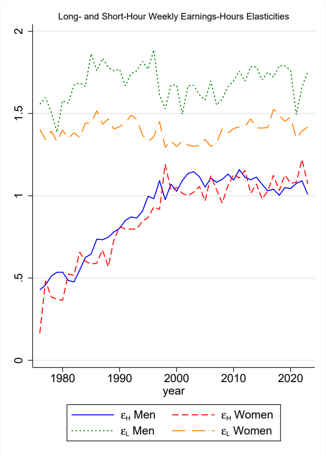
Separate long- and short-hour elasticities with controls | Possible Reasons | Counterfactual Exercises | Measurement Error | back

Allowing different elasticities for long and short hours

- Intuition: allow for different elasticities for "long" (> 40) and "short" (< 40) hours
- Specification:
  $\ln(Earnings) = \alpha + \varepsilon_\ell \mathbb{1}[\hat{h} < 0] + \varepsilon_h \mathbb{1}[\hat{h} > 0] + \gamma X$
  where X is a control vector, $\mathbb{1}(\cdot)$ is an indicator function, $\hat{h} = \ln(\frac{hrs}{40})$

  Back

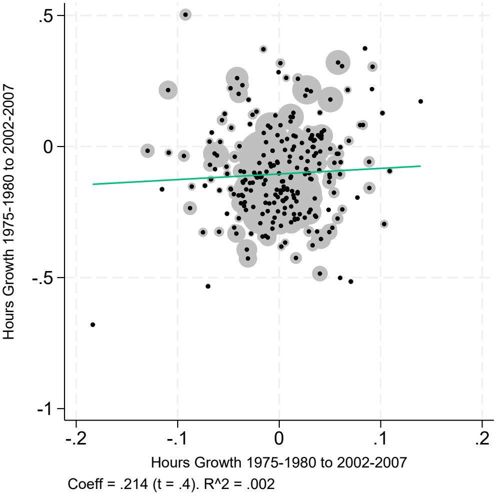
Medium-High and Low-Medium Elasticities



- Difference between men and women virtually disappears!

With Controls Back to main elasticities figure

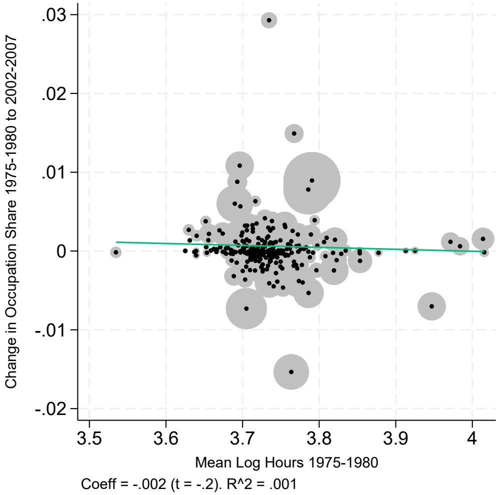
Hours growth and wage growth



- No relationship between hours growth and wage growth

back

“Early” hours and change in occupation share



- No relationship between hours and share change

back

Possible sources of change

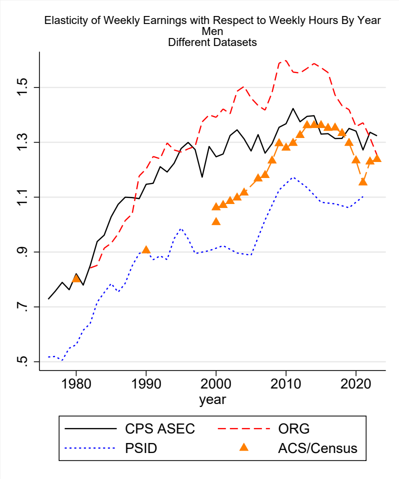
- Selection effects
  - Selection into long hours possibly changing across hours distribution
  - Return to long hours may be different across types of work (*e.g.* Gicheva, 2020)
- Could be change in hours distribution
  - Kuhn and Lozano (2008): men more likely to work long hours in 2000s relative to before
  - If elasticity is different at different hours levels, this could change elasticity estimates
- Change in hours-earnings technology
  - *E.g.*, technology could make longer work hours more valuable

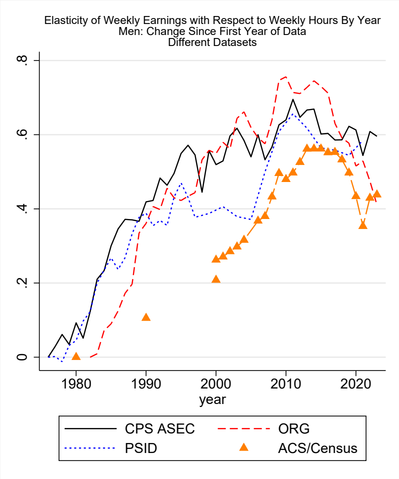
Back to main elasticities figure

Measurement error: Explanation

- Concern: hours are measured with error
- Measurement error in RHS variable → coefficient biased *downward*
- Key result here: elasticity has *increased over time*
  - Measurement error could explain result if measurement error *decreased over time*
  - Problem if, *e.g.*, CPS-ASEC's hours questions got better over time
- One simple check: do we see similar rise in other datasets?
- Other datasets with sufficient info: ACS/Census, PSID, CPS-ORG

Results from other datasets





Simple Counterfactual Exercise

- Let (h_{it}, e_{it}) be an observation of hours and earnings for a person i, from year t
- Regressions are estimated separately by t

Simple Counterfactual Exercise

- Let (h_{it}, e_{it}) be an observation of hours and earnings for a person i, from year t
- Regressions are estimated separately by t
- The elasticities are estimated from $\hat{\varepsilon}_t$ in the regression for a given t: $\ln(e_{it}) = \hat{\varepsilon}_t \ln(h_{it}) + \hat{\alpha}_t + \hat{u}_{it}$
  - Let $\hat{z}_{it} = \hat{\alpha}_t + \hat{u}_{it}$

Simple Counterfactual Exercise

- Let (h_{it}, e_{it}) be an observation of hours and earnings for a person i, from year t
- Regressions are estimated separately by t
- The elasticities are estimated from $\hat{\varepsilon}_t$ in the regression for a given t: $\ln(e_{it}) = \hat{\varepsilon}_t \ln(h_{it}) + \hat{\alpha}_t + \hat{u}_{it}$
  - Let $\hat{z}_{it} = \hat{\alpha}_t + \hat{u}_{it}$
- I estimate $\hat{\varepsilon}_{1976}$ from the regression for 1976

Simple Counterfactual Exercise

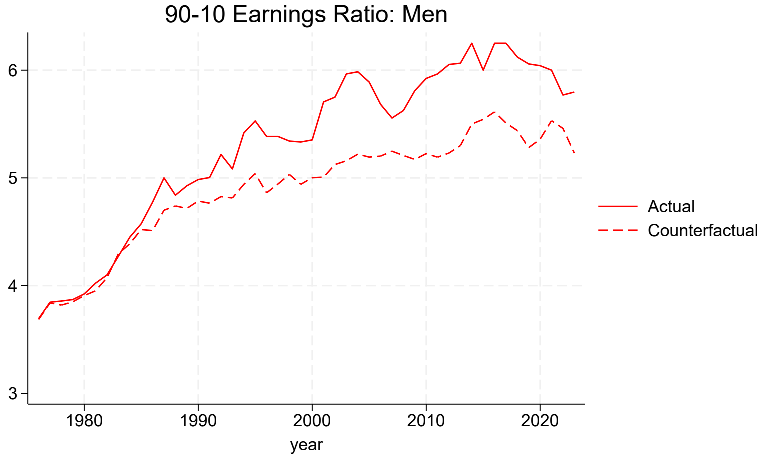
- Let (h_{it}, e_{it}) be an observation of hours and earnings for a person i, from year t
- Regressions are estimated separately by t
- The elasticities are estimated from $\hat{\varepsilon}_t$ in the regression for a given t: $\ln(e_{it}) = \hat{\varepsilon}_t \ln(h_{it}) + \hat{\alpha}_t + \hat{u}_{it}$
  - Let $\hat{z}_{it} = \hat{\alpha}_t + \hat{u}_{it}$
- I estimate $\hat{\varepsilon}_{1976}$ from the regression for 1976
- Then, compute "counterfactual" log earnings for year s as: $\ln(\hat{e}_{is}) = \hat{\varepsilon}_{1976} \ln(h_{is}) + \hat{z}_{is}$

Simple Counterfactual Exercise

- Let (h_{it}, e_{it}) be an observation of hours and earnings for a person i, from year t
- Regressions are estimated separately by t
- The elasticities are estimated from $\hat{\varepsilon}_t$ in the regression for a given t: $\ln(e_{it}) = \hat{\varepsilon}_t \ln(h_{it}) + \hat{\alpha}_t + \hat{u}_{it}$
  - Let $\hat{z}_{it} = \hat{\alpha}_t + \hat{u}_{it}$
- I estimate $\hat{\varepsilon}_{1976}$ from the regression for 1976
- Then, compute "counterfactual" log earnings for year s as: $\ln(\hat{e}_{is}) = \hat{\varepsilon}_{1976} \ln(h_{is}) + \hat{z}_{is}$
- Then, compute measures of earnings inequality for earnings (wage) and counterfactual earnings (wage)
  - 90th percentile-10th percentile earnings (wage) ratio
  - 75th percentile-25th percentile earnings (wage) ratio
  - Variance of log earnings

back

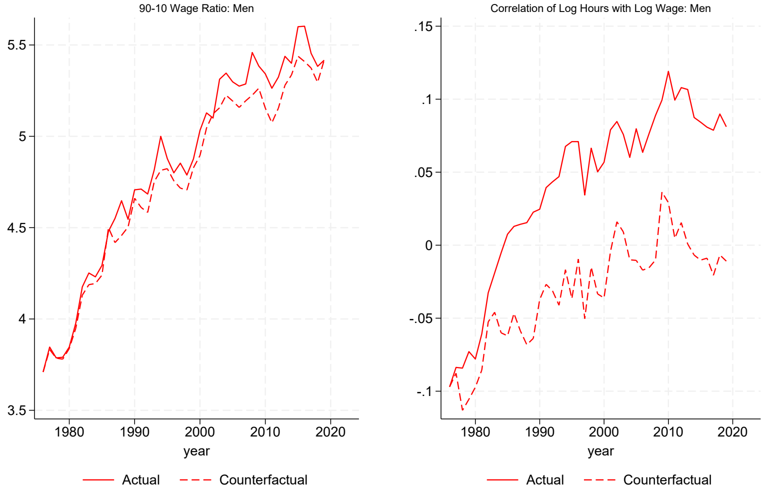
Actual and Counterfactual 90-10 Earnings Ratios



- Growth over full period in 90-10 ratio is roughly 22% lower in counterfactual

Variance of Log Earnings | Variance of Log Wage and Corr($\tilde{w}$, $\tilde{n}$) | 75-25 Ratio | back

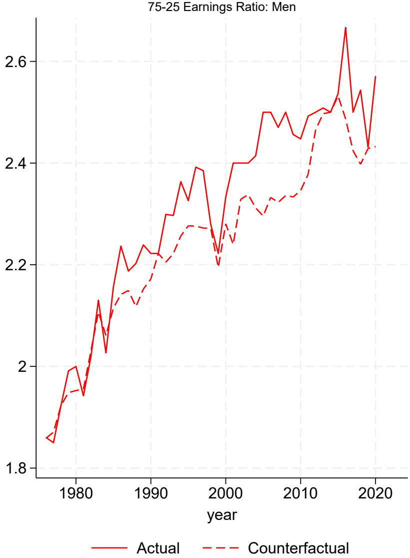
Wage and *corr*(*hours*, *wages*) counterfactuals



- Difference is from correlation between log hours and log wage

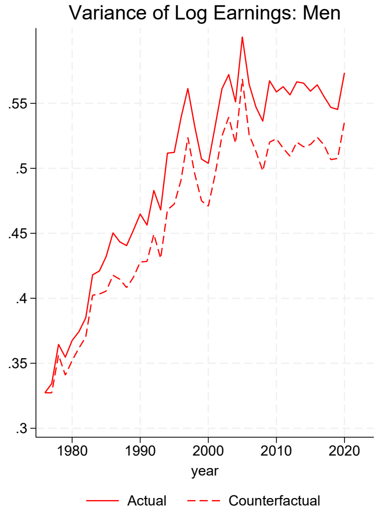
Back to Earnings 90-10

Actual and counterfactual 75-25 ratio



Back to Earnings 90-10

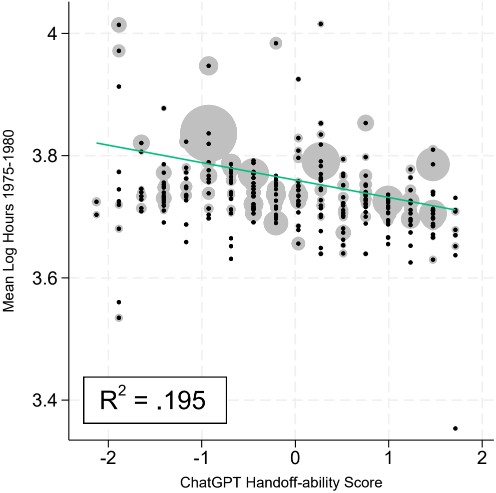
Actual and Counterfactual Variance of Log Earnings



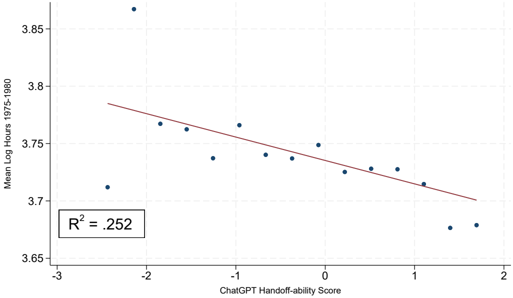
- Counterfactual variance is lower, especially from late 2000s on

Back to Earnings 90-10

Hours and Handoff-ability: Claude Score

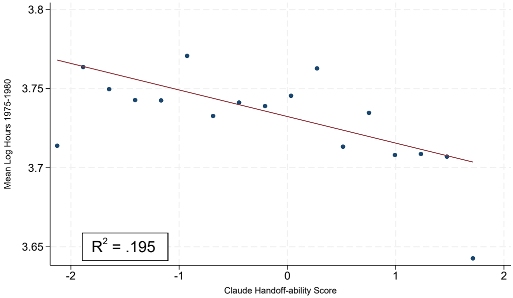


Hours and Handoff-ability: Binscatter



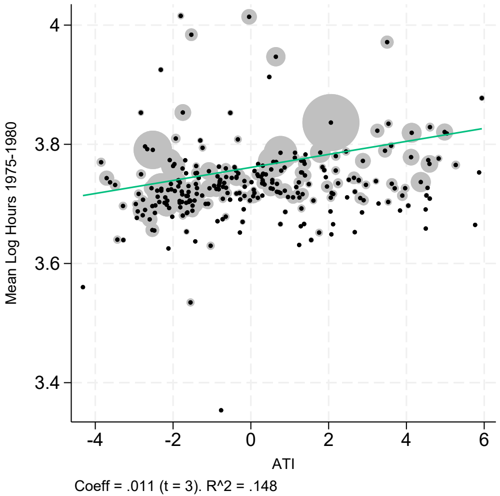
back

Hours and Handoff-ability: binscatter



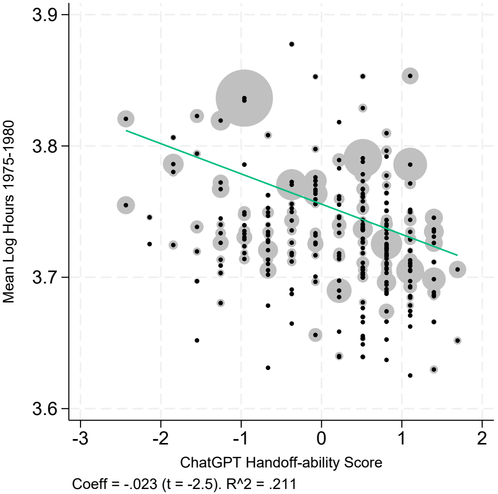
back

Hours and ATI



back

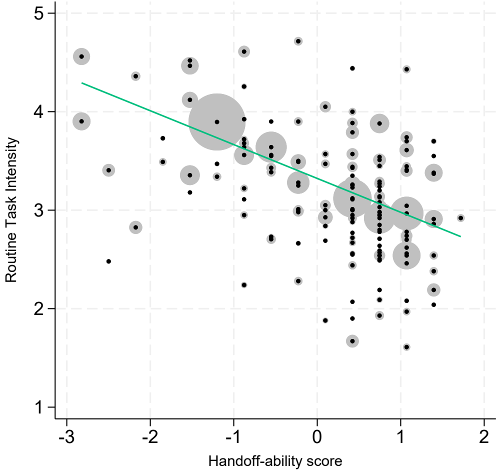
Hours and Handoff-ability: Values trimmed



- Drops values of mean log hours below 3.6 or above 3.9

back

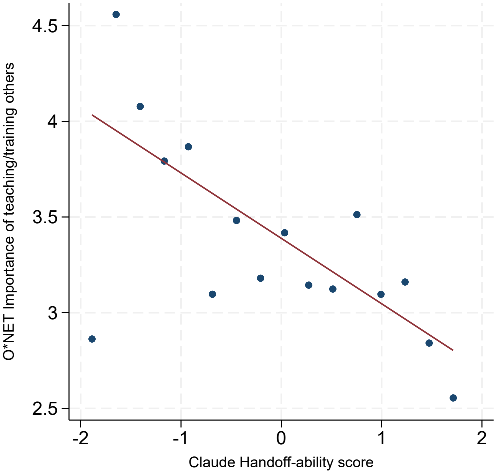
Importance of Documenting/Recording Information vs Handoff-ability



Coeff = -.344 (t = -6.7). R^2 = .447

back

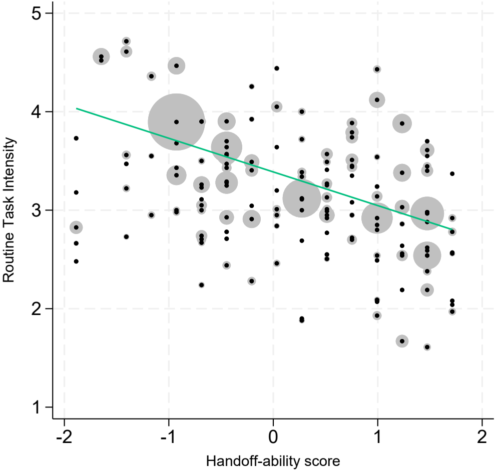
Importance of Documenting/Recording Information vs Handoff-ability



Coeff = -.342 (t = -5.2). R^2 = .35

back

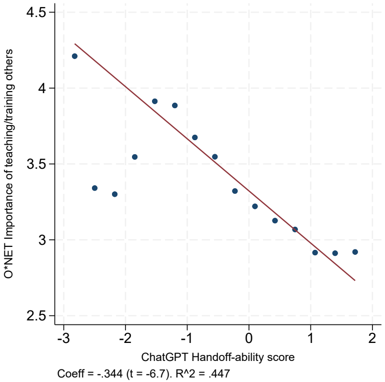
Importance of Documenting/Recording Information vs Handoff-ability



Coeff = -.342 (t = -5.2). R^2 = .35

back

Importance of Documenting/Recording Information vs Handoff-ability



- "Importance of documenting/recording information" from O*NET (proxy for c) decreasing in handoff-ability

Non-binscatter Claude Scores

LLM Measures

- Idea: ask LLMs to grade handoff-ability at the occupation level

  I am interested in "handoff-ability" as a characteristic of jobs, specifically how substitutable one worker is for another in production. I am interested specifically in how easily or well work in a job can be "handed off" to another worker. I want to upload an Excel sheet with a list of occupations, and I would like for you to grade each on this characteristic representing how well one worker can hand their work off to a coworker, from 1-100, in a new column called "score". Think specifically not of replacing a worker altogether, but of a coworker being able to pick up a person's work when, e.g., their shift ends. Do you understand these instructions? If so, can you give me a few clear examples of occupations with high handoff-ability and occupations with low handoff-ability?

LLM Measures

- Idea: ask LLMs to grade handoff-ability at the occupation level

  I am interested in "handoff-ability" as a characteristic of jobs, specifically how substitutable one worker is for another in production. I am interested specifically in how easily or well work in a job can be "handed off" to another worker. I want to upload an Excel sheet with a list of occupations, and I would like for you to grade each on this characteristic representing how well one worker can hand their work off to a coworker, from 1-100, in a new column called "score". Think specifically not of replacing a worker altogether, but of a coworker being able to pick up a person's work when, e.g., their shift ends. Do you understand these instructions? If so, can you give me a few clear examples of occupations with high handoff-ability and occupations with low handoff-ability?

- Correlation between Claude and ChatGPT scores ≈ 0.8